Exercise Sheet 9 - Solutions
Predicate Logic – Natural Deduction & Semantics

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1. Here is Natural Deduction proof of $(\forall x.\forall y.q(x,y) \to p(x)) \to \forall x.(\exists y.q(x,y)) \to p(x)$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\exists y.q(x,y)}\,{\scriptstyle 2} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\forall x.\forall y.q(x,y) \to p(x)}\,{\scriptstyle 1}}{\forall y.q(x,y) \to p(x)}\,[\forall E]}{q(x,y) \to p(x)}\,[\forall E] \qquad \dfrac{}{q(x,y)}\,{\scriptstyle 3}}{p(x)}\,[\to E]}{p(x)}\,{\scriptstyle 3}\,[\exists E]}{(\exists y.q(x,y)) \to p(x)}\,{\scriptstyle 2}\,[\to I]}{\forall x.(\exists y.q(x,y)) \to p(x)}\,[\forall I]}{(\forall x.\forall y.q(x,y) \to p(x)) \to \forall x.(\exists y.q(x,y)) \to p(x)}\,{\scriptstyle 1}\,[\to I]$$

2. Here is a Natural Deduction proof of $(\exists x.p(x)) \to (\forall x.\forall y.p(x) \to q(x,y)) \to (\exists x.\forall y.q(x,y))$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\exists x.p(x)}\,{\scriptstyle 1} \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\forall x.\forall y.p(x) \to q(x,y)}\,{\scriptstyle 2}}{\forall y.p(x) \to q(x,y)}\,[\forall E]}{p(x) \to q(x,y)}\,[\forall E] \qquad \dfrac{}{p(x)}\,{\scriptstyle 3}}{q(x,y)}\,[\to E]}{\forall y.q(x,y)}\,[\forall I]}{\exists x.\forall y.q(x,y)}\,[\exists I]}{\exists x.\forall y.q(x,y)}\,{\scriptstyle 3}\,[\exists E]}{(\forall x.\forall y.p(x) \to q(x,y)) \to (\exists x.\forall y.q(x,y))}\,{\scriptstyle 2}\,[\to I]}{(\exists x.p(x)) \to (\forall x.\forall y.p(x) \to q(x,y)) \to (\exists x.\forall y.q(x,y))}\,{\scriptstyle 1}\,[\to I]$$

3. Here is a proof of $A_1 \to A_2 \to A_3 \to A_4 \to C$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\Pi_1 \quad \Pi_2}{\texttt{pi1}(\texttt{pair}(y,x)) = y \to \texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = y}\,[\to E] \qquad \dfrac{\dfrac{\dfrac{}{A_3}\,{\scriptstyle 3}}{\forall z\texttt{pi1}(\texttt{pair}(y,z)) = y}\,[\forall E]}{\texttt{pi1}(\texttt{pair}(y,x)) = y}\,[\forall E]}{\texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = y}\,[\to E]}{\forall y.\texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = y}\,[\forall I]}{C}\,[\forall I]}{A_4 \to C}\,{\scriptstyle 4}\,[\to I]}{A_3 \to A_4 \to C}\,{\scriptstyle 3}\,[\to I]}{A_2 \to A_3 \to A_4 \to C}\,{\scriptstyle 2}\,[\to I]}{A_1 \to A_2 \to A_3 \to A_4 \to C}\,{\scriptstyle 1}\,[\to I]$$

where Π_1 is

$$\dfrac{\dfrac{\dfrac{\dfrac{}{A_4}\,{\scriptstyle 4}}{\forall w.\forall z.\texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = w \to w = z \to \texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = z}\,[\forall E]}{\forall z.\texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = \texttt{pi1}(\texttt{pair}(y,x)) \to \texttt{pi1}(\texttt{pair}(y,x)) = z \to \texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = z}\,[\forall E]}{\texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = \texttt{pi1}(\texttt{pair}(y,x)) \to \texttt{pi1}(\texttt{pair}(y,x)) = y \to \texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = y}\,[\forall E]$$

where Π_2 is

$$\dfrac{\dfrac{\dfrac{\overline{A_2}^{\;2}}{\texttt{swap}(\texttt{pair}(x,y)) = z \rightarrow \texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = \texttt{pi1}(z)}\,[\forall E]}{\texttt{swap}(\texttt{pair}(x,y)) = \texttt{pair}(y,x) \rightarrow \texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = \texttt{pi1}(\texttt{pair}(y,x))}\,[\forall E] \qquad \dfrac{\dfrac{\overline{A_1}^{\;1}}{\forall y.\texttt{swap}(\texttt{pair}(x,y)) = \texttt{pair}(y,x)}\,[\forall E]}{\texttt{swap}(\texttt{pair}(x,y)) = \texttt{pair}(y,x)}\,[\forall E]}{\texttt{pi1}(\texttt{swap}(\texttt{pair}(x,y))) = \texttt{pi1}(\texttt{pair}(y,x))}\,[\rightarrow E]$$

4.
- $\vDash_{M,\cdot} \forall x.\texttt{even}(x) \rightarrow \exists y.\texttt{odd}(y) \wedge y > x$
- iff for all $n \in \mathbb{N}$, $\vDash_{M,x\mapsto n} \texttt{even}(x) \rightarrow \exists y.\texttt{odd}(y) \wedge y > x$
- iff for all $n \in \mathbb{N}$, if $\vDash_{M,x\mapsto n} \texttt{even}(x)$ then $\vDash_{M,x\mapsto n} \exists y.\texttt{odd}(y) \wedge y > x$
- iff for all $n \in \mathbb{N}$, if $\langle [\![x]\!]^M_{x\mapsto n}\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$ then $\vDash_{M,x\mapsto n} \exists y.\texttt{odd}(y) \wedge y > x$
- iff for all $n \in \mathbb{N}$, if $\langle n\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$ then $\vDash_{M,x\mapsto n} \exists y.\texttt{odd}(y) \wedge y > x$
- iff for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then $\vDash_{M,x\mapsto n} \exists y.\texttt{odd}(y) \wedge y > x$
- iff for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then there exists a $m \in \mathbb{N}$ such that $\vDash_{M,x\mapsto n,y\mapsto m} \texttt{odd}(y) \wedge y > x$
- iff for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then there exists a $m \in \mathbb{N}$ such that $\vDash_{M,x\mapsto n,y\mapsto m} \texttt{odd}(y)$ and $\vDash_{M,x\mapsto n,y\mapsto m} y > x$
- iff for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then there exists a $m \in \mathbb{N}$ such that $\langle [\![y]\!]^M_{x\mapsto n,y\mapsto m}\rangle \in \{\langle 1\rangle, \langle 3\rangle, \langle 5\rangle, \dots\}$ and $\langle [\![y]\!]^M_{x\mapsto n,y\mapsto m}, [\![x]\!]^M_{x\mapsto n,y\mapsto m}\rangle \in \{\langle 1,0\rangle, \langle 2,0\rangle, \langle 2,1\rangle, \dots\}$
- iff for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then there exists a $m \in \mathbb{N}$ such that $\langle m\rangle \in \{\langle 1\rangle, \langle 3\rangle, \langle 5\rangle, \dots\}$ and $\langle m, n\rangle \in \{\langle 1,0\rangle, \langle 2,0\rangle, \langle 2,1\rangle, \dots\}$
- iff for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then there exists a $m \in \mathbb{N}$ such that $m \in \{1, 3, 5, \dots\}$ and $\langle m, n\rangle \in \{\langle 1,0\rangle, \langle 2,0\rangle, \langle 2,1\rangle, \dots\}$
- True, because given $n \in \{0, 2, 4, \dots\}$, we can pick m to be $n+1$, which is in $\{1, 3, 5, \dots\}$, and which also satisfy $\langle n+1, n\rangle \in \{\langle 1,0\rangle, \langle 2,0\rangle, \langle 2,1\rangle, \dots\}$

5.
- $\neg \vDash_{M,\cdot} \forall x.\texttt{even}(x) \rightarrow \texttt{even}(\texttt{succ}(x))$
- iff it is not true that for all $n \in \mathbb{N}$, $\vDash_{M,x\mapsto n} \texttt{even}(x) \rightarrow \texttt{even}(\texttt{succ}(x))$
- iff it is not true that for all $n \in \mathbb{N}$, if $\vDash_{M,x\mapsto n} \texttt{even}(x)$ then $\vDash_{M,x\mapsto n} \texttt{even}(\texttt{succ}(x))$
- iff it is not true that for all $n \in \mathbb{N}$, if $\langle [\![x]\!]^M_{x\mapsto n}\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$ then $\vDash_{M,x\mapsto n} \texttt{even}(\texttt{succ}(x))$
- iff it is not true that for all $n \in \mathbb{N}$, if $\langle n\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$ then $\vDash_{M,x\mapsto n} \texttt{even}(\texttt{succ}(x))$
- iff it is not true that for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then $\vDash_{M,x\mapsto n} \texttt{even}(\texttt{succ}(x))$
- iff it is not true that for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then $\langle [\![\texttt{succ}(x)]\!]^M_{x\mapsto n}\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$
- iff it is not true that for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then $\langle +1([\![x]\!]^M_{x\mapsto n})\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$
- iff it is not true that for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then $\langle +1(n)\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$
- iff it is not true that for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then $\langle n+1\rangle \in \{\langle 0\rangle, \langle 2\rangle, \langle 4\rangle, \dots\}$
- iff it is not true that for all $n \in \mathbb{N}$, if $n \in \{0, 2, 4, \dots\}$ then $n+1 \in \{0, 2, 4, \dots\}$
- True, because for example $0+1$ is not in $\{0, 2, 4, \dots\}$

6. M' can for example be
- $\langle \mathbb{N}, \langle 0, \langle x\rangle \mapsto x\rangle, \langle \{\langle x\rangle \mid x \text{ is odd}\}, \{\langle x\rangle \mid x \text{ is odd}\}, \emptyset\rangle\rangle$
- $\langle \mathbb{N}, \langle 0, \langle x\rangle \mapsto x{+}2\rangle, \langle \{\langle x\rangle \mid x \text{ is even}\}, \{\langle x\rangle \mid x \text{ is odd}\}, \emptyset\rangle\rangle$
- $\langle \mathbb{N}, \langle 0, \langle x\rangle \mapsto x{+}1\rangle, \langle \emptyset, \{\langle x\rangle \mid x \text{ is odd}\}, \emptyset\rangle\rangle$